UTS SISTEM BASIS DATA

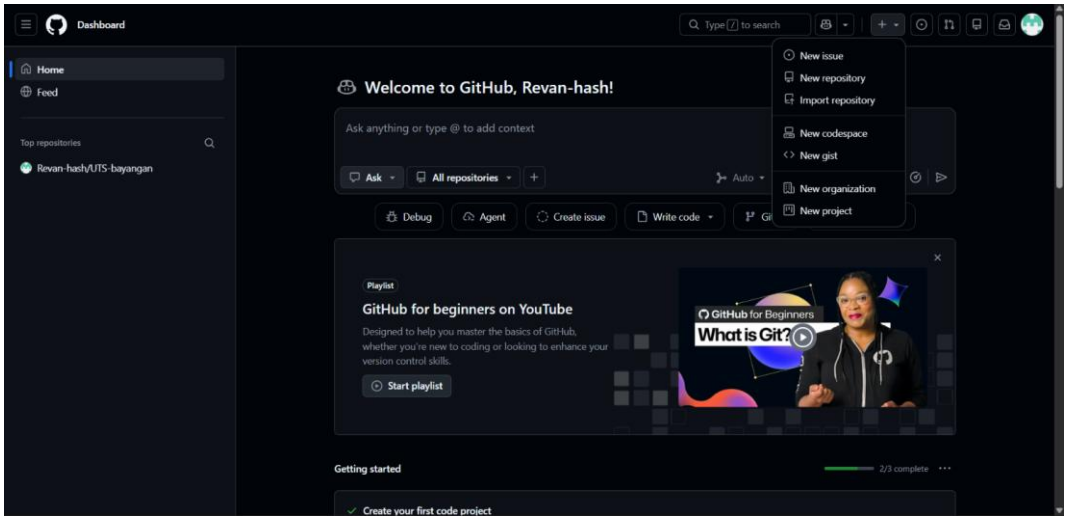
MUHAMMAD REVAN ADRIAN HADINATA

1251420109

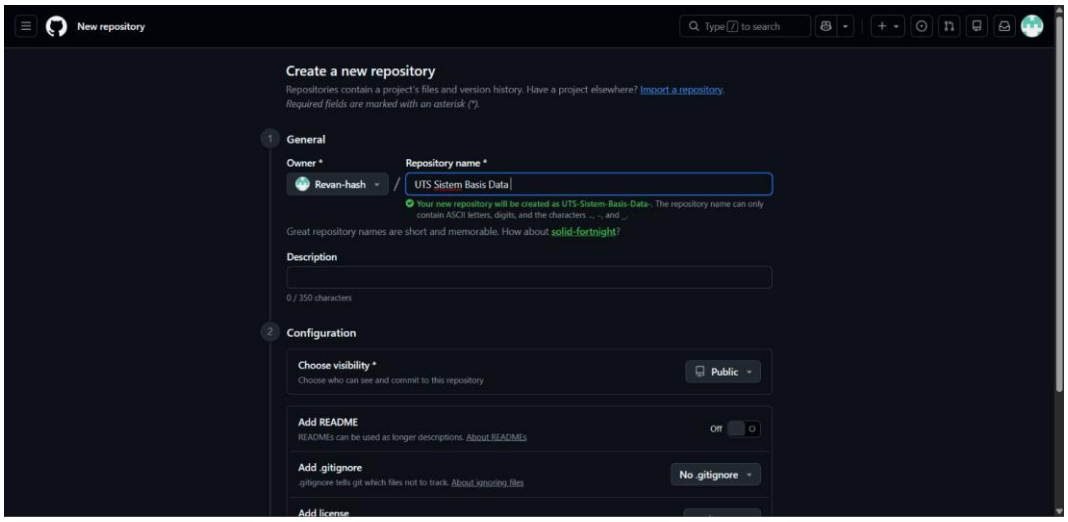
3D

A.STEP BY STEP GITHUB

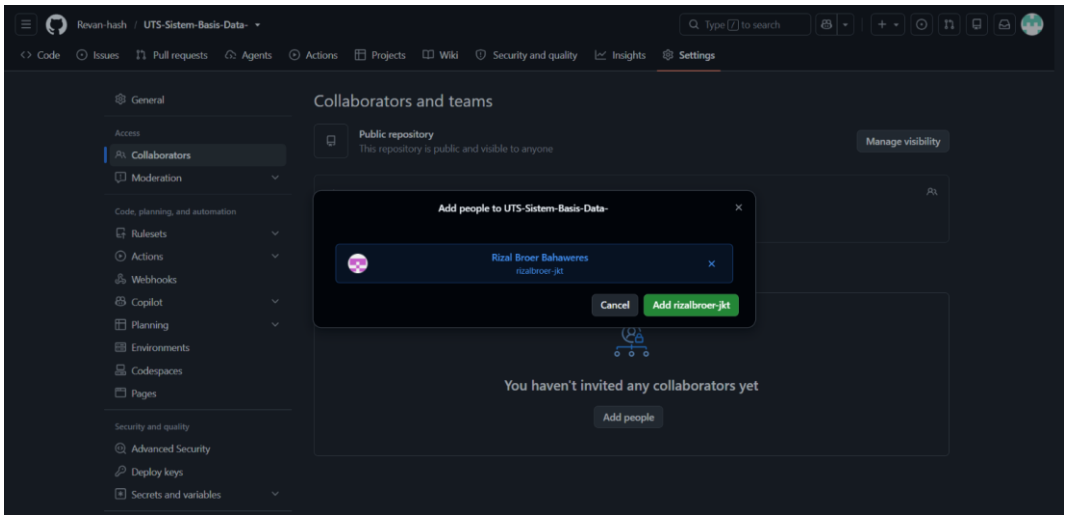
1.Membuat repository



2.Menamakan repository



3.Undang collaborator Pak Broer



B.STEP BY STEP MEMBUAT QUERRY SQL

SOAL 3.21

1. Langkah 1: Membuat Skrip DDL Logical Model (Soal 3.21)

1. Buka Text Editor, buat file baru bernama soal_3_21_logical.sql.
2. Masukkan skrip SQL berikut:

```sql
CREATE TABLE STATE (
    State_name VARCHAR(50) NOT NULL,
    Region VARCHAR(20) NOT NULL,
    CONSTRAINT pk_state PRIMARY KEY (State_name),
    CONSTRAINT chk_region CHECK (Region IN ('Northeast', 'Midwest', 'Southeast', 'Southwest', 'West'))
);

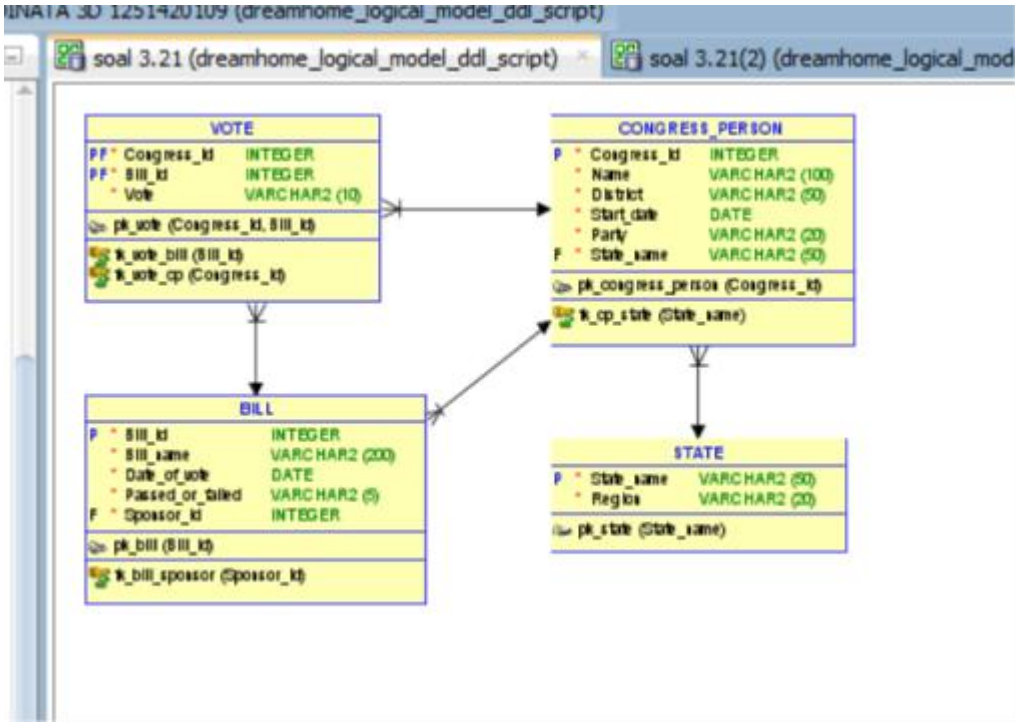
CREATE TABLE CONGRESS_PERSON (
    Congress_id INT NOT NULL,
    Name VARCHAR(100) NOT NULL,
    District VARCHAR(50) NOT NULL,
    Start_date DATE NOT NULL,
    Party VARCHAR(20) NOT NULL,
    State_name VARCHAR(50) NOT NULL,
    CONSTRAINT pk_congress_person PRIMARY KEY (Congress_id),
    CONSTRAINT fk_cp_state FOREIGN KEY (State_name) REFERENCES STATE(State_name),
    CONSTRAINT chk_party CHECK (Party IN ('Republican', 'Democrat', 'Independent', 'Other'))
);

CREATE TABLE BILL (
    Bill_id INT NOT NULL,
    Bill_name VARCHAR(200) NOT NULL,
    Date_of_vote DATE NOT NULL,
    Passed_or_failed VARCHAR(5) NOT NULL,
    Sponsor_id INT NOT NULL,
    CONSTRAINT pk_bill PRIMARY KEY (Bill_id),
    CONSTRAINT fk_bill_sponsor FOREIGN KEY (Sponsor_id) REFERENCES CONGRESS_PERSON(Congress_id),
    CONSTRAINT chk_passed CHECK (Passed_or_failed IN ('Yes', 'No'))
);
```

```sql
CREATE TABLE VOTE (
    Congress_id INT NOT NULL,
    Bill_id INT NOT NULL,
    Vote VARCHAR(10) NOT NULL,
    CONSTRAINT pk_vote PRIMARY KEY (Congress_id, Bill_id),
    CONSTRAINT fk_vote_cp FOREIGN KEY (Congress_id) REFERENCES CONGRESS_PERSON(Congress_id),
    CONSTRAINT fk_vote_bill FOREIGN KEY (Bill_id) REFERENCES BILL(Bill_id),
    CONSTRAINT chk_vote_val CHECK (Vote IN ('Yes', 'No', 'Abstain', 'Absent'))
);
```

Langkah 2: Impor ke Oracle Data Modeler & Generate Logical Model

1. Buka Oracle SQL Developer Data Modeler.
2. Klik menu File → Import → DDL File.
3. Klik tombol Add, pilih file soal_3_21_logical.sql, lalu klik OK.
4. Pada jendela Database Sites, pilih Oracle Database 21c, lalu klik OK.
5. Setelah muncul jendela *View Log* (Status: 0 Failed), klik Close.
6. Pada jendela *Compare Models*, klik Merge (atau Close).
7. Klik kanan di area putih lembar kerja diagram → Auto Layout → All untuk merapikan tampilan diagram.



SOAL 3.34

Langkah 1: Membuat Skrip DDL Logical Model (Soal 3.34)

1. Buat file baru bernama soal_3_34_logical.sql.
2. Masukkan skrip SQL berikut:

```sql
CREATE TABLE AUTHOR (
    Email VARCHAR(100) NOT NULL,
    First_name VARCHAR(50) NOT NULL,
    Last_name VARCHAR(50) NOT NULL,
    CONSTRAINT pk_author PRIMARY KEY (Email)
);

CREATE TABLE PAPER (
    Paper_id INT NOT NULL,
    Title VARCHAR(200) NOT NULL,
    Abstract TEXT,
    File_name VARCHAR(100) NOT NULL,
    Contact_author_email VARCHAR(100) NOT NULL,
    CONSTRAINT pk_paper PRIMARY KEY (Paper_id),
    CONSTRAINT fk_paper_contact_author FOREIGN KEY (Contact_author_email) REFERENCES AUTHOR(Email)
);

CREATE TABLE PAPER_AUTHOR (
    Paper_id INT NOT NULL,
    Author_email VARCHAR(100) NOT NULL,
    CONSTRAINT pk_paper_author PRIMARY KEY (Paper_id, Author_email),
    CONSTRAINT fk_pa_paper FOREIGN KEY (Paper_id) REFERENCES PAPER(Paper_id),
    CONSTRAINT fk_pa_author FOREIGN KEY (Author_email) REFERENCES AUTHOR(Email)
);

CREATE TABLE REVIEWER (
    Email VARCHAR(100) NOT NULL,
    First_name VARCHAR(50) NOT NULL,
    Last_name VARCHAR(50) NOT NULL,
    Phone_number VARCHAR(20),
    Affiliation VARCHAR(100),
    CONSTRAINT pk_reviewer PRIMARY KEY (Email)
);
```

```sql
CREATE TABLE REVIEWER_TOPIC (
    Reviewer_email VARCHAR(100) NOT NULL,
    Topic VARCHAR(100) NOT NULL,
    CONSTRAINT pk_reviewer_topic PRIMARY KEY (Reviewer_email, Topic),
    CONSTRAINT fk_rt_reviewer FOREIGN KEY (Reviewer_email) REFERENCES REVIEWER(Email)
);

CREATE TABLE REVIEW (
    Review_id INT NOT NULL,
    Paper_id INT NOT NULL,
    Reviewer_email VARCHAR(100) NOT NULL,
    Technical_merit INT CHECK (Technical_merit BETWEEN 1 AND 10),
    Readability INT CHECK (Readability BETWEEN 1 AND 10),
    Originality INT CHECK (Originality BETWEEN 1 AND 10),
    Relevance INT CHECK (Relevance BETWEEN 1 AND 10),
    Overall_recommendation VARCHAR(50),
    Committee_comment TEXT,
    Author_feedback TEXT,
    CONSTRAINT pk_review PRIMARY KEY (Review_id),
    CONSTRAINT fk_rev_paper FOREIGN KEY (Paper_id) REFERENCES PAPER(Paper_id),
    CONSTRAINT fk_rev_reviewer FOREIGN KEY (Reviewer_email) REFERENCES REVIEWER(Email)
```

Langkah 2: Impor ke Oracle Data Modeler & Logical Diagram

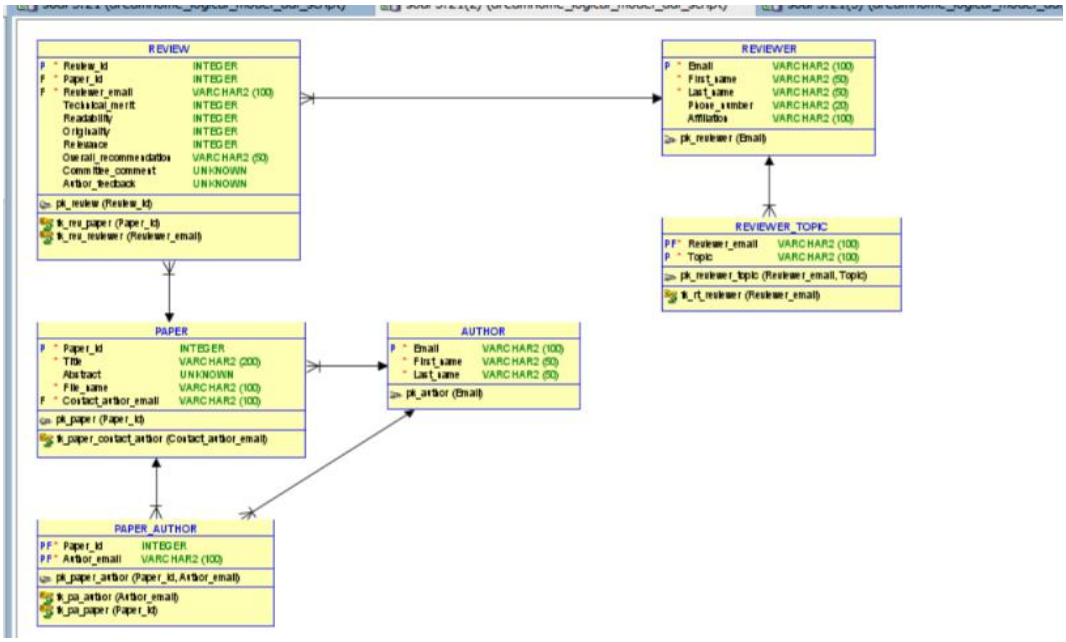
1. Di Oracle Data Modeler: Klik **File → Import → DDL File**.
2. Pilih file soal_3_34_logical.sql, pilih **Oracle Database 21c → OK**.
3. Selesaikan jendela *Log* dan *Compare Models* (klik **Merge** / **Close**).
4. Klik kanan canvas → **Auto Layout → All**.
5. **Pengujian Langkah:** Pastikan entitas AUTHOR, PAPER, PAPER_AUTHOR, REVIEWER, REVIEWER_TOPIC, dan REVIEW tampil lengkap.

Langkah 3: Physical Model & Export DDL PostgreSQL

1. Tekan **Ctrl + F4** → Klik **Engineer**.
2. Klik **File → Export → DDL File**.
3. Simpan file dengan nama soal_3_34_postgres.sql.

Langkah 4: Dump ke PostgreSQL

1. Di pgAdmin, buat Database baru: db_conference_review.
2. Buka **Query Tool** pada db_conference_review.
3. Tempel isi file soal_3_34_postgres.sql, lalu tekan **F5**.
4. **Pengujian Langkah:** Pastikan seluruh 6 tabel dan *check constraints* berhasil dieksekusi.

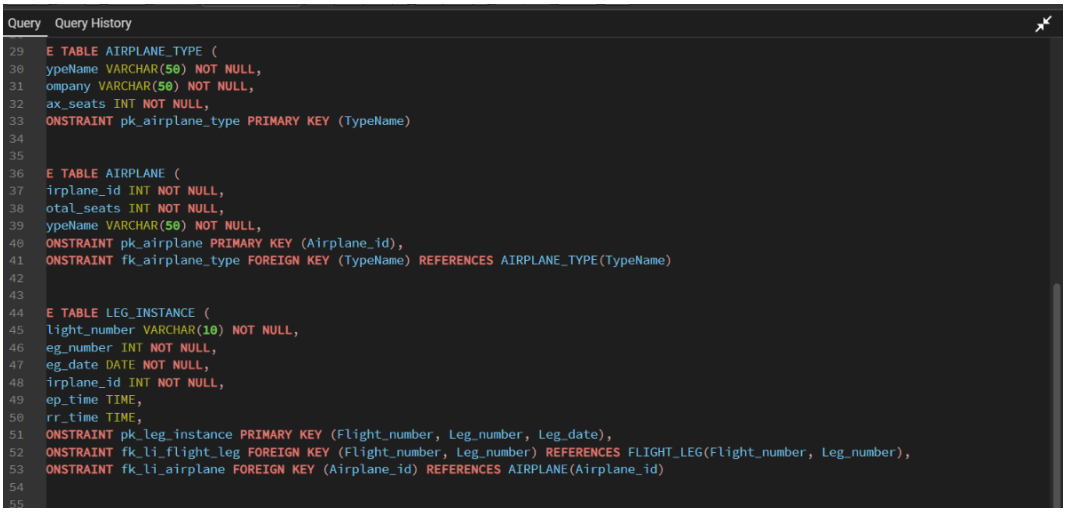


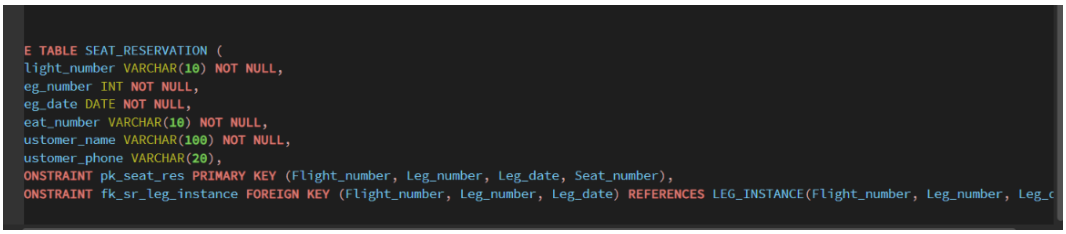
SOAL 3.35

Langkah 1: Membuat Skrip DDL Logical Model (Soal 3.35)

1. Buat file baru bernama soal_3_35_logical.sql.
2. Masukkan skrip SQL berikut:

```
Query  Query History
1   E TABLE AIRPORT (
2   irport_code VARCHAR(10) NOT NULL,
3   ame VARCHAR(100) NOT NULL,
4   ity VARCHAR(50) NOT NULL,
5   tate VARCHAR(50) NOT NULL,
6   ONSTRAINT pk_airport PRIMARY KEY (Airport_code)
7
8
9   E TABLE FLIGHT (
10  light_number VARCHAR(10) NOT NULL,
11  irline VARCHAR(50) NOT NULL,
12  eekdays VARCHAR(50),
13  ONSTRAINT pk_flight PRIMARY KEY (Flight_number)
14
15
16  E TABLE FLIGHT_LEG (
17  light_number VARCHAR(10) NOT NULL,
18  eg_number INT NOT NULL,
19  eparture_airport_code VARCHAR(10) NOT NULL,
20  rrival_airport_code VARCHAR(10) NOT NULL,
21  cheduled_dep_time TIME,
22  cheduled_arr_time TIME,
23  ONSTRAINT pk_flight_leg PRIMARY KEY (Flight_number, Leg_number),
24  ONSTRAINT fk_leg_flight FOREIGN KEY (Flight_number) REFERENCES FLIGHT(Flight_number),
25  ONSTRAINT fk_leg_dep_airport FOREIGN KEY (Departure_airport_code) REFERENCES AIRPORT(Airport_code),
26  ONSTRAINT fk_leg_arr_airport FOREIGN KEY (Arrival_airport_code) REFERENCES AIRPORT(Airport_code)
27
28
```





Langkah 2: Impor ke Oracle Data Modeler & Logical Diagram

1. Pilih **File → Import → DDL File**.
2. Impor berkas soal_3_35_logical.sql, pilih site **Oracle Database 21c → OK**.
3. Selesaikan proses *import* dan klik **Merge / Close**.
4. Klik kanan canvas → **Auto Layout → All**.
5. **Pengujian Langkah:** Ambil *screenshot* diagram Logical Model yang menampilkan entitas AIRPORT, FLIGHT, FLIGHT_LEG, AIRPLANE_TYPE, AIRPLANE, LEG_INSTANCE, dan SEAT_RESERVATION.

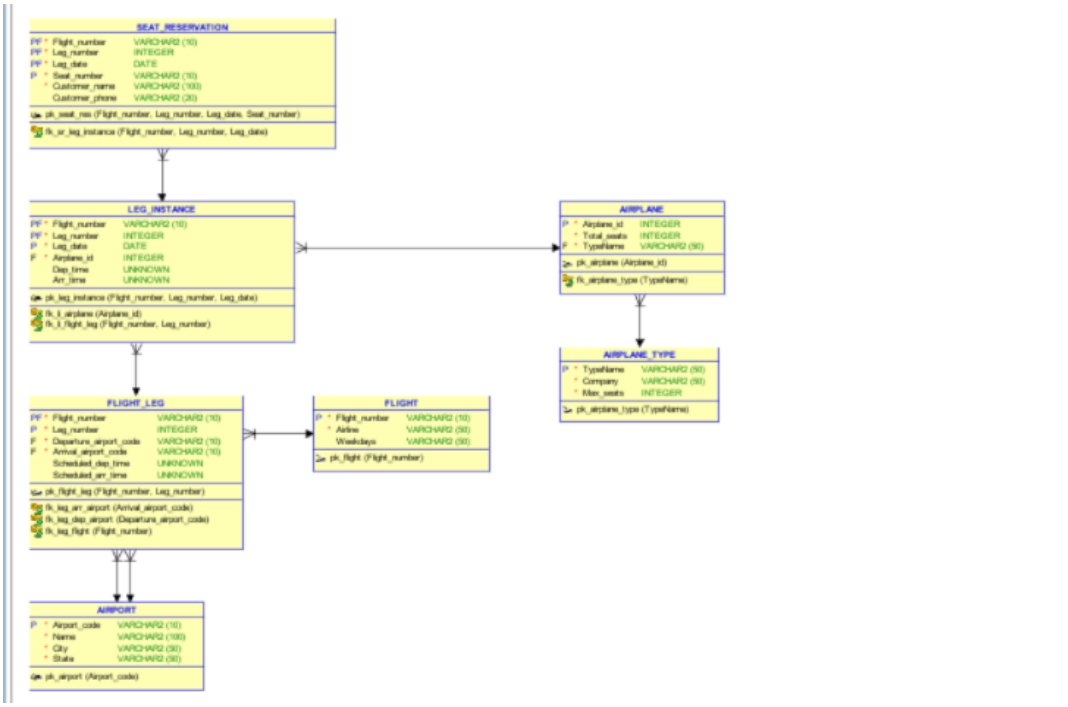
Langkah 3: Physical Model & Export DDL PostgreSQL

1. Tekan **Ctrl + F4** → Klik **Engineer**.
2. Ekspor DDL melalui **File → Export → DDL File**.
3. Simpan file dengan nama soal_3_35_postgres.sql.

Langkah 4: Dump ke PostgreSQL

1. Di pgAdmin, buat Database baru: db_airline.
2. Buka **Query Tool** pada database db_airline.
3. Tempel isi berkas soal_3_35_postgres.sql dan jalankan (**F5**).

4. **Pengujian Langkah:** Pastikan seluruh 7 tabel dan foreign key terhubung dengan sukses tanpa pesan kesalahan.



DAN SELESAI